2025-2026 CSE320 COMPUTER NETWORKS RECITATION

Q1. Fill in the Space

I. The distance vector algorithm is ______ and ______ in nature. (iterative and asynchronous)
II. A node updates its distance vector when there is a change in ______ or when it receives a message from a ______. (local link cost, neighbor)
III. In distance vector routing, nodes notify neighbors only when their ______ changes. (Distance Vector)
IV. The technique used to mitigate routing loops in distance vector is called ______ ______. (poison reverse)
V. In BGP, ______ BGP is used to obtain subnet reachability information from neighboring ASes. (eBGP)
VI. In ARP, the broadcast MAC address used for queries is ______.
VII. In physical layer communication, ______ refers to random energy added to the signal.
VIII. AS numbers (ASNs) are typically ______-bit values.
IX. The process of changing a bit stream before transmission is called ______.
X. Ethernet's MAC protocol uses ______ with binary backoff.

Q2. Compare

Explain one difference in each comparison

I. Compare **iterative** and **asynchronous** behavior in routing algorithms.
    A. Iterative refers to repeated computation steps, while asynchronous means updates occur
II. Compare **intra-domain routing** and **inter-domain routing**.
    A. Intra-domain routing operates within an AS, while inter-domain routing operates between
III. Compare **MAC address** and **IP address**.
    A. MAC address is a flat physical address used locally, while IP address is hierarchical and used for global routing.
IV. Compare **encoding** and **modulation**.
    A. Encoding converts bits into signal patterns, while modulation alters signal properties like frequency or amplitude.
V. Compare **connectionless** and **reliable** communication in Ethernet.
    A. Ethernet is connectionless and does not establish a session, and it is unreliable because it does not provide acknowledgments.

Q3. HTTP Page Load

I. Non-persistent

A student opens:

`http://www.ceng.edu.tr/home.html`

Assume:

- Browser has no cached content.
- DNS resolution requires **3 RTTs**.
- Server uses **non-persistent HTTP/1.0**.
- The base HTML file contains **5 embedded objects**.
- All objects are on the **same server**.
- Each TCP connection setup requires **1 RTT**.
- Each HTTP request/response requires **1 RTT**.
- File transmission times are neglected.
- Non-persistent HTTP opens a **new TCP connection for each object**.

How many RTTs are required until the full page is received? **A. 15 RTT**

II. Persistent

A student opens:

`http://www.networks.edu.tr/page.html`

Assume:

- DNS resolution requires **2 RTTs**.
- Server uses **persistent HTTP/1.1 without pipelining**.
- The base HTML file contains **4 embedded objects**.
- All objects are on the **same server**.
- TCP connection setup requires **1 RTT**.
- Each HTTP request/response requires **1 RTT**.
- The same TCP connection remains open.
- File transmission time is neglected.

How many RTTs are required? **A. 8 RTT**

III. Persistent HTTP with pipelining

A student opens:

`http://www.lab.edu.tr/main.html`

Assume:

- DNS resolution requires **2 RTTs**.
- Server uses **persistent HTTP with pipelining**.
- The base HTML file contains **6 embedded objects**.
- All objects are on the **same server**.
- TCP setup requires **1 RTT**.
- Base HTML request/response requires **1 RTT**.
- After receiving the HTML, the browser sends all embedded object requests together.
- All embedded objects are received after **1 RTT**.
- File transmission times are neglected.

How many RTTs are required?

A. 5 RTT

IV. Mixed servers, non-persistent HTTP

A student opens:

`http://www.cs.metu.edu.tr/index.html`

Assume:

- DNS for the main server requires **2 RTTs**.
- DNS for an external image server requires **1 RTT**.
- Server uses **non-persistent HTTP**.
- Base HTML is on `www.cs.metu.edu.tr`.
- The HTML contains:
  - **3 objects** from the main server
  - **2 objects** from `cdn.example.com`
- Each TCP connection setup requires **1 RTT**.
- Each HTTP request/response requires **1 RTT**.
- File transmission times are neglected.

How many RTTs are required? **A. 15 RTT**

V. Non-persistent HTTP with parallel connections

A student opens:

`http://www.final.edu.tr/home.html`

Assume:

- DNS resolution requires **2 RTTs**.
- Server uses **non-persistent HTTP**.
- Base HTML contains **8 embedded objects**.
- All objects are on the same server.
- TCP setup requires **1 RTT**.
- HTTP request/response requires **1 RTT**.
- Browser can open **4 parallel TCP connections**.
- File transmission times are neglected.

How many RTTs are required?

A. **8RTT**

Q4. TCP RTT and Timeout Estimation

I.

A TCP sender measures the following SampleRTTs in milliseconds:

$$[110,\ 120,\ 115,\ 130,\ 100]$$

Initial values:

$$EstimatedRTT = 100$$

$$DevRTT = 5$$

Use:

$$EstimatedRTT = (1-\alpha)EstimatedRTT + \alpha SampleRTT$$

$$DevRTT = (1-\beta)DevRTT + \beta|SampleRTT - EstimatedRTT|$$

$$TimeoutInterval = EstimatedRTT + 4DevRTT$$

Assume:

$$\alpha = 0.125, \quad \beta = 0.25$$

Calculate the final **TimeoutInterval** after all 5 samples.

Answer Step-by-Step:

SampleRTT	EstimatedRTT	DevRTT	TimeoutInterval
110	101.25	5.94	125.00
120	103.59	8.55	137.81
115	105.02	8.91	140.66
130	108.14	12.15	156.73
100	107.12	10.89	150.69

Final:

$$TimeoutInterval = 107.12 + 4(10.89)$$

$$TimeoutInterval = 150.69\ ms$$

Final answer: 150.69 ms

II. Next

A TCP sender measures the following SampleRTTs (in milliseconds):

$$[90,\ 100,\ 105,\ 95,\ 110]$$

Initial values:

$$EstimatedRTT = 80$$

$$DevRTT = 10$$

Use:

$$EstimatedRTT = (1 - \alpha)EstimatedRTT + \alpha SampleRTT$$

$$DevRTT = (1 - \beta)DevRTT + \beta|SampleRTT - EstimatedRTT|$$

$$TimeoutInterval = EstimatedRTT + 4DevRTT$$

Assume:

$$\alpha = 0.125, \quad \beta = 0.25$$

Answer Step-by-Step:

SampleRTT	EstimatedRTT	DevRTT	TimeoutInterval
90	81.25	9.69	120.00
100	83.59	11.37	129.07
105	86.27	12.96	138.11
95	87.36	11.63	133.88
110	90.19	13.67	144.87

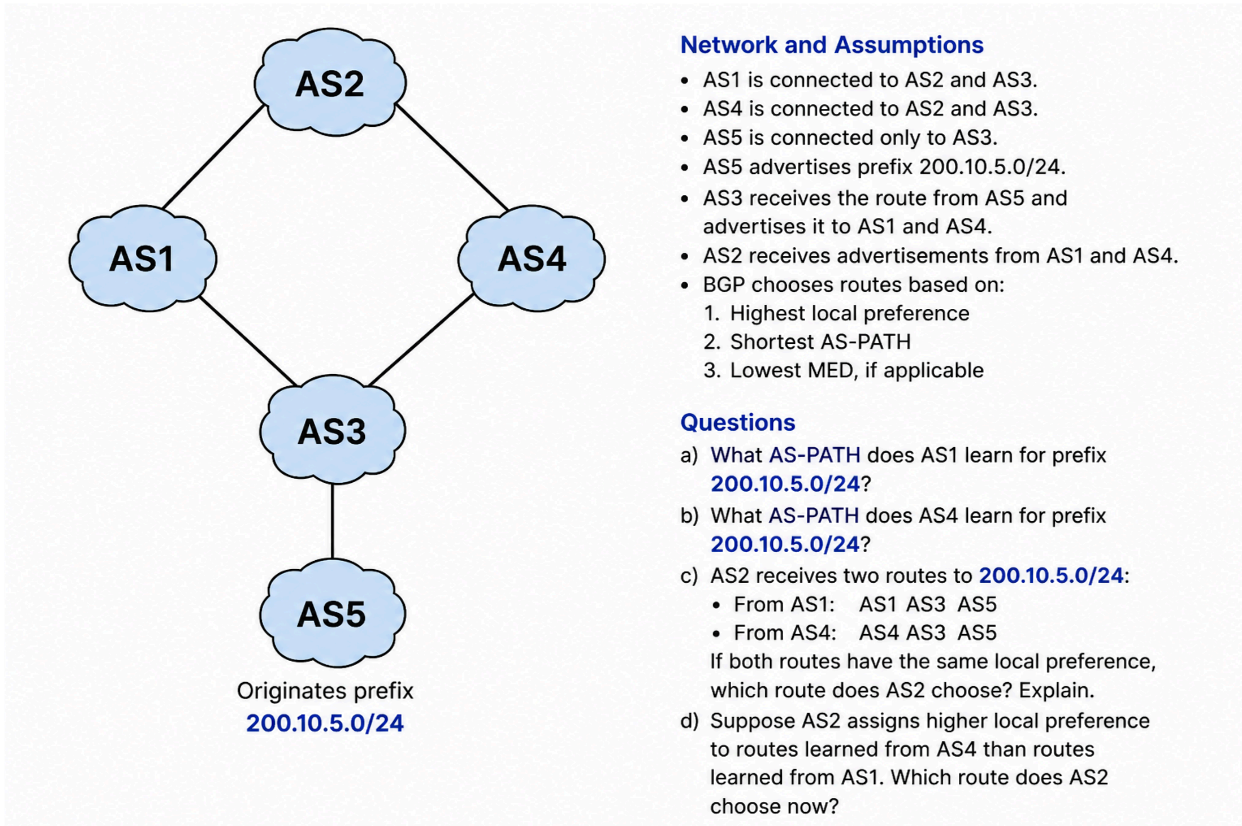
Final calculation:

$$TimeoutInterval = 90.19 + 4(13.67)$$

$$TimeoutInterval = 144.87\ ms$$

Q5. BGP

I. Network Ex1



a) What AS-PATH does AS1 learn for prefix **200.10.5.0/24**?

Answer: AS1 learns the route:AS3 AS5

Because AS3 advertises the prefix learned from AS5 to AS1.

b) What AS-PATH does AS4 learn for prefix **200.10.5.0/24**?

Answer: AS4 learns the route: AS3 AS5

Because AS4 also receives the route through AS3.

c) AS2 receives two routes to **200.10.5.0/24**:

- From AS1: `AS1 AS3 AS5`
- From AS4: `AS4 AS3 AS5`

If both routes have the same local preference, which route does AS2 choose? Explain.

Answer:

AS2 compares:

AS1 AS3 AS5 and

AS4 AS3 AS5

Both AS-PATHs have length 3.

Since local preference and AS-PATH length are equal, **<u>AS2 cannot choose based only on these two criteria</u>**. It must use the next BGP tie-breaker, such as MED or another rule.

d) Suppose AS2 assigns higher local preference to routes learned from AS4 than routes learned from AS1. Which route does AS2 choose now?

Answer:

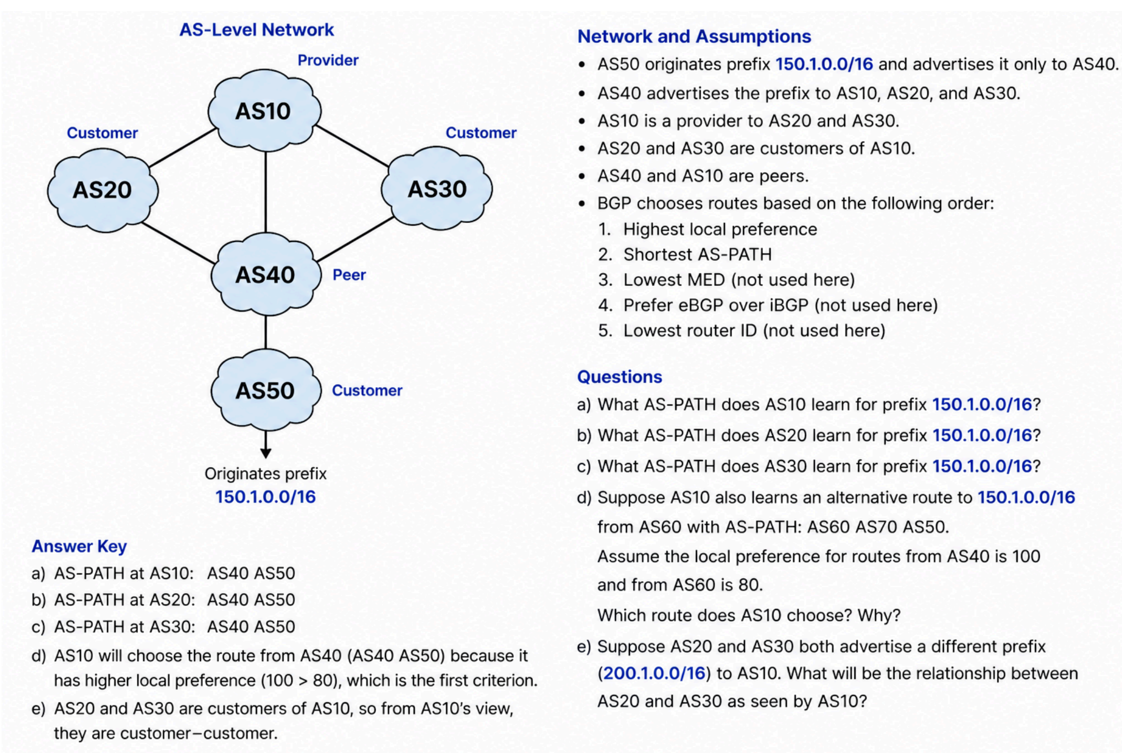
AS2 chooses the route through AS4:

AS4 AS3 AS5

Because BGP prefers the route with the **highest local preference** before comparing AS-PATH length.

Final answer: AS2 selects the route via **AS4**.

II. Network Ex2



Q6. TCP Congestion and Retransmission

You are running a file upload service. A user reports that their uploads start fast but then slow down dramatically. You inspect Wireshark logs and notice multiple duplicate ACKs and retransmissions after a few seconds of smooth transfer.

Questions:

a. What TCP mechanism is likely causing the slowdown?
b. Explain how TCP interprets duplicate ACKs.
c. Which part of the TCP algorithm kicks in, and how does it behave?

Q7. DNS Resolution Problem in Local Network

A user complains they can access websites by IP address (like `http://142.250.190.78`) but not by domain name (like `http://www.google.com`). Their device has a correct IP and can ping external IPs.

Network setup:

- The DNS server is configured manually as `192.168.1.1`
- That IP belongs to the user's **home router**
- All other network settings (IP, gateway, subnet) are valid.

Question:

a. What layer is likely malfunctioning here?
b. What protocol is responsible for resolving domain names?
c. Suggest two ways to troubleshoot or solve the issue.